It is automatic that $0 \leq \Pr(\mathcal{A}) \leq 1$. Certain properties need to be assumed regarding ρ and the way that this is formalized in mathematics is by description of the vector space of functions to which ρ belongs.

A useful class of functions from a domain $\mathcal{D} \subset \mathbb{R}^m$ to $\mathbb{R}$ are those that satisfy the following boundedness condition:

$$\int_{\mathcal{D}} \phi^2(z) \mathrm{d}\omega < \infty. \tag{5.3}$$

The set of functions from $\mathcal{D} \subset \mathbb{R}^m$ to $\mathbb{R}$ which satisfy (5.3) is referred to as the space $L^2(\mathcal{D})$ and we define the $L^2(\mathcal{D})$ norm by

$$\|\phi\|_{L^2(\mathcal{D})} \stackrel{\text{def}}{=} \left[\int_{\mathcal{D}} \phi^2(z) \mathrm{d}\omega \right]^{1/2}.$$

This is a real *Hilbert space* or *complete inner product space* in which, for any $\phi, \psi \in L^2(\mathcal{D})$, the inner product is defined by

$$\langle \phi, \psi \rangle_{L^2(\mathcal{D})} \stackrel{\text{def}}{=} \int_{\mathcal{D}} \phi(z)\psi(z) \mathrm{d}\omega.$$

Thus $\|\phi\|^2_{L^2(\mathcal{D})} = \langle \phi, \phi \rangle_{L^2(\mathcal{D})}$.

The inner product is a symmetric, bilinear operation: if α, β are real scalars and $f, g, h \in L^2(\mathcal{D})$, then

$$\langle \alpha f + \beta g, h \rangle = \alpha \langle f, h \rangle + \beta \langle g, h \rangle,$$

and

$$\langle f, g \rangle = \langle g, f \rangle.$$

The space $L^2(\mathbb{R}^m)$ includes for example the continuous functions with compact (closed, bounded) support, where the *support* of a function f is the closure of the set of arguments for which the function is nonzero and is denoted supp$f$.[2]

[2] $L^2(\mathbb{R}^m)$ also includes many other types of functions important in molecular modelling, for example the multivariate Gaussian $\exp(-\sum_{i=1}^m \alpha_i x_i^2)$, where $\alpha_i > 0$, $i = 1, 2, \ldots, m$ are coefficients; such a function goes rapidly to zero as any $x_i \to \infty$, so the improper integral converges. $L^2(\mathbb{R}^m)$ also includes the compactly supported functions with finite "jump discontinuities" at a finite number of locations.